

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

System Scope, Functional, and Non-Functional Requirements for VBCUA

1. System Scope & Objectives

The VBCUA is a desktop/web-based assessment engine designed to capture student audio responses, transcribe them locally, check semantic similarity against a stored reference concept, analyze acoustic parameters, and compile a standardized evaluation. The primary goal is to provide a zero-latency, local, and secure evaluation environment.

2. Functional Requirements

- ****FR-1 (Audio Ingestion)****: The system shall accept WAV and MP3 audio files via a file uploader, validating duration and file size.
- ****FR-2 (Acoustic Feature Extraction)****: The system shall parse the raw audio file using Librosa to calculate duration, Root-Mean-Square (RMS) energy, and the proportion of silence intervals (pause ratio).
- ****FR-3 (Local Transcription)****: The system shall transcribes the audio using OpenAI Whisper, bypassing external ffmpeg executable command-line installations on Windows by feeding pre-resampled mono numpy arrays directly into the Whisper decoder.
- ****FR-4 (Semantic Evaluation)****: The system shall use the SentenceTransformer 'all-MiniLM-L6-v2' model to calculate the cosine similarity between the transcription and the reference concept.
- ****FR-5 (Fluency Analytics)****: The system shall scan the transcribed text using regex boundaries to count filler words ('um', 'uh', 'like', 'ah', 'eh', 'so', 'actually', 'basically', 'you know') and compute a filler ratio.
- ****FR-6 (Database Records)****: The system shall log all evaluations, users, sessions, audio features, similarity metrics, and report links to a local relational SQLite database using SQLAlchemy.
- ****FR-7 (Report Compilation)****: The system shall generate a styled PDF report showing the score cards, text content comparison, and diagnostic recommendation notes.

3. Non-Functional Requirements

- ****NFR-1 (Accuracy)****: The Whisper transcription must maintain a low Word Error Rate (WER) on clear english recordings. The Sentence-BERT semantic similarity must map to standard human grading metrics.
- ****NFR-2 (Latency)****: Subsequent evaluations of cached models must execute in under 3 seconds.
- ****NFR-3 (Data Security)****: All transcriptions and evaluations must be executed locally on the user's system; no audio files or texts shall be transmitted to external third-party APIs.

- **NFR-4 (Portability)**: The application shall run on any Windows system containing Python 3.10+ and the required packages.